

Coding & Solution

Date	03 July 2026
Team ID	XXXXXX
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented Smart Lender application and the overall machine learning solution. The project demonstrates working functionality, modular code structure, clean coding practices, and successful implementation of loan eligibility prediction using machine learning. The application is developed using Python and Flask and deployed on the Render cloud platform.

Instructions:

1. Ensure the code is well-structured, properly commented, and follows best coding practices.
2. The implemented solution addresses the defined problem statement and satisfies the functional requirements of the Smart Lender project.
3. Include all source files, machine learning model files, configuration files, dataset, and setup instructions required to execute the application.
4. Provide the GitHub repository link for source code reference.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/CHANDRASAI9491/Smart-Lender-Loan-Eligibility-Prediction
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost
Key Features Implemented	• Loan eligibility prediction using Machine Learning. • Data preprocessing and feature encoding. • Model training using Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost. • XGBoost selected as the final deployed model. • Flask-based web application. • Responsive user interface for entering applicant details. • Real-time loan eligibility prediction. • Trained model integration using <code>model.pkl</code> and <code>scaler.pkl</code>. • Input validation and prediction result display. • Deployment on Render cloud platform.
Pending / Incomplete Features	• Database integration. • User authentication. • Prediction history storage. • Loan application management dashboard. • Email/SMS notification service.

Setup / Run Instructions

1. Clone the GitHub repository.

git clone <https://github.com/CHANDRASAI9491/Smart-Lender-Loan-Eligibility-Prediction.git>

2. Open the project folder.

3. Install the required Python libraries.

pip install -r requirements.txt

4. Navigate to the Flask application folder.

cd flask

5. Run the Flask application.

python app.py

6. Open the browser.

<http://127.0.0.1:5000>

7. Enter the applicant details.

8. Click Predict.

9. View the loan eligibility prediction result.

Live Deployment

<https://smart-lender-loan-eligibility-prediction-8448.onrender.com>

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into reusable functions and modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments and documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository (GitHub)	Yes

Additional Notes / Comments:

The **Smart Lender – Loan Eligibility Prediction Using Machine Learning** project is developed using **Python, Flask, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, and Seaborn**. The application predicts loan eligibility based on applicant information by utilizing a trained **XGBoost** model. The project follows a modular architecture with separate components for data preprocessing, machine learning model training, Flask web application development, and deployment on the **Render** cloud platform. The source code is maintained in a GitHub repository, making it easy to maintain, enhance, and extend for future development.